# An Empirical Analysis of Transfer for Dynamic Path Planning

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## Abstract

We study the dynamic path planning problem, where an agent needs to plan a route in the presence of moving obstacles. This problem is relevant for all agents that need to navigate in the real world. In particular, we study whether heuristics learned in training maps can be transferred—zero-shot or from a few labeled target maps—to new environments for this task. We show that, with the right training objective and planner integration, a frozen heuristic transfers zero-shot across the procedural continuous dynamic suites—even given no information about future obstacle motion—improving success over a geometric anchor while substantially reducing search effort; transfer does not extend to externally authored MovingAI-derived geometry, and selected discrete-grid experiments show a smaller rank-only effect. With a few labeled target maps, low-rank adaptation preserves the transferred prior while full fine-tuning overtakes it as supervision spans more target worlds—and a matched-label factorial shows that the coverage of distinct target worlds, not raw label count, predicts when full fine-tuning preserves held-out success. Further, across comparable backbones no architecture is uniformly best: Multilayer Perceptrons, Long Short-Term Memory models, Hierarchical Reasoning Models and U-Net architectures can all support this transfer, while hierarchy-specific gains do not persist across the tested formulations.

## Introduction

Dynamic path planning searches over both configuration and time: when obstacles move, a planner must consider waiting, detouring, and timing, and the search space grows accordingly (Silver 2005; Phillips and Likhachev 2011). The geometric admissible heuristics that make A* (Hart, Nilsson, and Raphael 1968) efficient on static maps ignore exactly these temporal effects, so search effort grows sharply in the presence of moving obstacles. Learned heuristics can reduce search effort (Arfaee, Zilles, and Holte 2011; Bhardwaj, Choudhury, and Scherer 2017; Yonetani et al. 2021), but a deployed planner benefits only if a heuristic trained on one set of maps remains useful on maps it has never seen. We study zero-shot and few-shot transfer of learned heuristics to unseen environments for dynamic path planning.

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We organize the experiments around three factors—the training formulation, the way the prediction enters the planner, and the distribution of target supervision—each evaluated first on static suites, where supervision is exact, then under deterministic dynamics. Comparisons use matched controls throughout—twin models differing in one input, identical weights under different integrations, anchor-calibrated budgets, map-level paired inference (Henderson et al. 2018; Agarwal et al. 2021; Pineau et al. 2021)—and confirmatory comparisons, units, and criteria were specified before test evaluation. Across the tested settings, representation, planner integration, and supervision regime explain the observed transfer outcomes more consistently than model family.

Our contributions are:

- **Dynamic zero-shot transfer with one fixed learned heuristic.** A single U-Net heuristic—development-selected, frozen before the confirmation cohort was generated, and given no future-motion input—improves success on all six dynamic suites of a 50-map-per-suite confirmation cohort ($+0.18$ to $+0.84$; every paired map-level CI excludes zero) and cuts matched search effort by 73–95%. Three suites are held out (a distinct topology; corridor/patroller and scale shifts); the rest are training families on disjoint maps. Against per-suite tuned weighted A* it keeps one BH-significant success advantage (spiral) and lower search effort with intervals excluding parity on four of six suites, at a significant path-quality cost on two; an optimal safe-interval planner solves 0.98–1.00 of the same instances outright, locating the learned advantage within budgeted time-expanded search; two independent retrainings with larger collections reproduce the success pattern.
- **Planner integration changes whether transferred predictions reduce search.** Changing the input–target–output formulation rescues a collapsed recurrent model; changing only the planner integration turns an inert discrete residual into useful guidance; and sharing one search state turns an exact oracle from a six-map loss into a six-map win. These controls show that a negative planning result does not isolate the model unless the supervised target and search interface are held fixed.
- **Distinct-world coverage controls adaptation stability in the matched-label factorial.** With scarce static su-

pervision, low-rank adaptation preserves the transferred prior while full fine-tuning is unstable; with more supervision, full fine-tuning often overtakes it (direct map-paired contrasts at both ends). A matched-label factorial ties the low-supervision failure to concentration in too few distinct worlds, not raw label count: distributing the same labels across more worlds rescues full fine-tuning, and independent world-draw replicates reproduce the coverage contrasts. Low-rank adaptation is consistently the most stable method, though one replicate shows its protection is not absolute.

Additional experiments bound the scope: future-window inputs, adapter interpolation, and hierarchy-specific depth add no consistent value, while a restricted-observation heuristic with one Bellman backup beats a complete-map comparator on expansions.

## Problem Formulation and Learned-Heuristic Interfaces

**Planning problems.** In the continuous setting, a point robot plans on a probabilistic roadmap (PRM; Kavraki et al. 1996) built over a square world of side length $L$=1.0 (2.0 for the large-room family); units are abstract. Each roadmap has 192 nodes in total—the start and goal at indices 0 and 1 plus 190 sampled free-space nodes—connected by $k$=7 nearest-neighbor candidate edges per node (doubled for the start and goal, then symmetrized into an undirected graph); edges are validated by sampled collision checks at resolution $0.01L$, and worlds whose roadmap does not connect start to goal are rejected. Static search states are roadmap nodes, edge costs are Euclidean lengths $\ell(e)$, and the anchor heuristic is Euclidean distance to the goal, $h_0(v) = \|x_v - x_g\|$. In the dynamic setting, obstacles move on deterministic trajectories and the search state is a pair $(v, t)$ of node and discrete time step of duration $\Delta t$. The planner may traverse an edge, taking $\tau(e) = \lceil \ell(e)/(v_{\mathrm{agent}}\Delta t)\rceil$ steps, or wait one step; transitions are validated against the obstacle trajectories by sampled collision checks along the edge and interval. Cost is arrival time in steps, search runs to a horizon $t_{\max}$, and the anchor is the travel-time lower bound $h_0(v, t) = \|x_v - x_g\|/(v_{\mathrm{agent}}\Delta t)$, constant in $t$, so anchor and cost share units. In the discrete setting, agents plan on occupancy grids of 20×20 to 256×256 cells with Manhattan distance as the anchor.

**Supervision targets.** With $h^*$ the exact cost-to-go (graph Dijkstra on the static roadmap; backward space–time Dijkstra over $(v, t)$ in the dynamic setting), the regression target is the normalized anchor residual $\bar{\delta}^*(s) = \mathrm{clip}(h^*(s) - h_0(s),\, 0,\, cT)\,/T$, where $T$ is the normalization scale ($T$=$L$ statically; $T = L/(v_{\mathrm{agent}}\Delta t)$, the map-crossing time in steps, dynamically—the same value for every suite by construction) and the cap is $c$=4.0 in normalized units, applied after normalization; unreachable states are masked from the loss. Scalar models predict $\bar{\delta}_\theta$ from per-node feature tokens; field models predict a goal-conditioned residual raster (eight input channels; appendix) sampled at node positions. At integration the prediction is rescaled: $\hat{h}_\theta(s) = h_0(s) + T \cdot \mathrm{clip}(\bar{\delta}_\theta(s), 0, c)$.

**Planner integrations.** Under *additive* integration the planner orders states by $f(s) = g(s) + \hat{h}_\theta(s)$, sacrificing admissibility for guidance; empirical path-cost ratios are reported wherever additive integration is used. The classical control is weighted A* (Pohl 1970), which orders states by $f(s) = g(s) + w_h\, h_0(s)$ with a per-suite tuned inflation $w_h$. Under *focal (rank-only)* integration (Pohl 1970; Pearl and Kim 1982), the anchor priority $f_0(s) = g(s) + h_0(s)$ defines the band $\mathrm{FOCAL}_w = \{s \in \mathrm{OPEN} : f_0(s) \le w \cdot \min_{u \in \mathrm{OPEN}} f_0(u)\}$, and the learned prediction orders expansions only within the band. At $w$=1.0 the learned score merely breaks ties among minimum-priority states, preserving the anchor's path-cost optimality when search completes (finite budgets can still change success). In the restricted-observation study, learned predictions serve as exit values of a radius-bounded local subgraph; one inference-time Bellman backup over that subgraph produces the heuristic, with a scale factor calibrated once on a disjoint block.

**Transfer protocol.** Zero-shot transfer applies a trained-and-fixed heuristic to unseen maps and unseen map families with no target supervision. Few-shot adaptation draws $K$ labeled target maps and compares low-rank adaptation (LoRA; Hu et al. 2022) (rank 8 for the HRM, rank 24 for the wider ON-LSTM), full fine-tuning, and training from scratch: LoRA and full fine-tuning start from the same trained checkpoint; scratch uses the same architecture and schedule from random initialization. (In the compositional-mission study, $K$ denotes mission length; flagged where used.)

## Experimental Setup

**Environment families.** A *family* is a procedural map generator; a *suite* is a named evaluation condition; a *world* is one sampled environment. Continuous worlds come from our procedural generator (specification tables in the appendix; code in the anonymized artifact archive accompanying the submission). The static hard families are wall constructions—*maze/dense maze* (corridor mazes; the dense variant narrows gaps to $0.115L$), *rooms/large rooms* (partition walls with doorways; the large variant doubles the side length), *spiral*, and *bugtrap* (exact generator rules in the appendix). Static training uses the maze, rooms, and spiral families; dense maze, bugtrap, and large rooms are held out from training (bugtrap is a distinct topology; dense maze and large rooms are gap-width- and scale-shifted variants of trained generators). Dynamic suites add 3–5 moving circular obstacles per world that sweep line segments with triangle-wave motion (radius 0.055–$0.080L$, span 0.36–$0.50L$, period 0.14–0.30 of the horizon; per-suite parameters in the appendix), plus a *crossing* open-arena suite as a timing control. *Dynamic training draws worlds only from the dynamic maze, rooms, and spiral families; dynamic dense maze, crossing, and large rooms are held out*—crossing as a distinct open-arena topology, dense maze with narrower corridors and adjusted patroller parameters, large rooms at doubled scale with a faster agent—so three of the six dynamic test suites share a generator with training (evaluated on disjoint maps) and three are distribution-shifted at the suite level. No learned model observes a test world during training; eval-

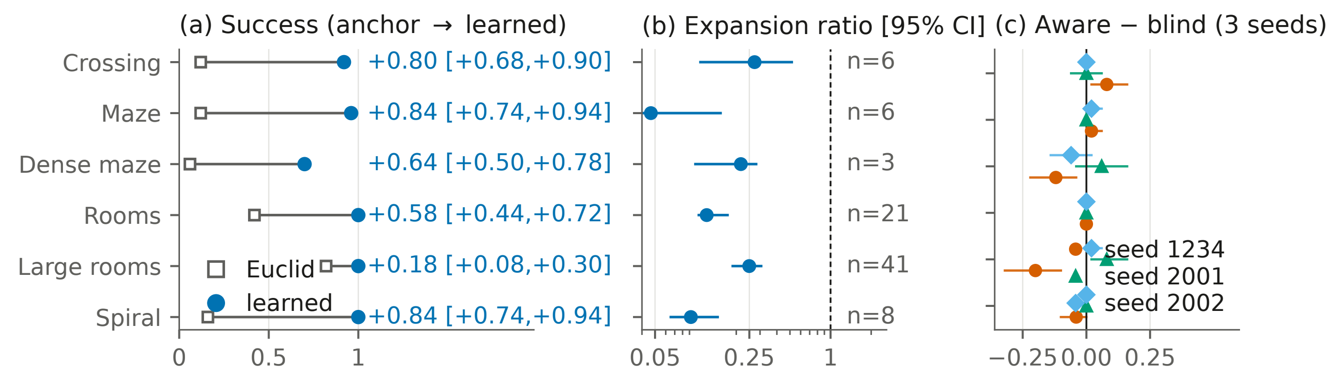


Figure 1: Dynamic zero-shot transfer of one fixed blind U-Net heuristic (blind = no future-motion input) on the 50-map-per-suite confirmation cohort (crossing is a held-out topology; dense maze and large rooms are held-out parameter/scale shifts of trained generators; the other three are trained families on disjoint maps). (a) Anchor (open squares) and learned (filled circles) success per suite; labels: paired mean delta with map-level 95% CI (all six exclude zero; marginal). (b) Matched-solved expansion ratios (log axis, map-level 95% CIs); $n$ = jointly solved count (dense maze $n$=3 descriptive). (c) Aware−blind success differences for the three training pipelines; the large-rooms cell flips sign. Details in the appendix.

uation cohorts come from disjoint seed ranges, checked by world fingerprints.

**Models.** Scalar models consume per-node token sequences (62 tokens of 16 dimensions; appendix) and include an MLP, an LSTM (Hochreiter and Schmidhuber 1997), an ON-LSTM (Shen et al. 2019) (1,252,481 parameters), and a Hierarchical Reasoning Model (HRM; Wang et al. 2025) (2,158,529 parameters) with an adaptive-computation port (Graves 2016; Banino, Balaguer, and Blundell 2021). Field models consume the eight-channel $64\times64$ raster and include a FiLM-conditioned U-Net (Ronneberger, Fischer, and Brox 2015; Perez et al. 2018) (3,702,370 parameters), a field HRM (4,016,642), and a field ON-LSTM (1,448,578); dynamic variants append $W$ future-occupancy channels ($W$=8 aware, giving ground-truth occupancy at offsets $t$+1, …, $t$+8; $W$=0 blind). Unless a specific experiment states otherwise, the continuous residual models train with AdamW (learning rate $2\times10^{-4}$, weight decay $10^{-4}$, batch 256, gradient clipping 1.0) on a smooth-L1 loss over the capped normalized residual; complete architecture and optimizer tables, including the additional MLP, LSTM, and graph-network configurations used by the mission experiments, are in the appendix. Within each direct architecture comparison, models are trained under one matched recipe on identical data.

**Evaluation.** Each suite is evaluated at a per-suite expansion budget selected by anchor-only calibration: candidate budgets are swept over a fixed grid, and the grid budget whose anchor success is closest to the pre-specified calibration targets (evenly spaced in $[0.45, 0.70]$; ties to the smaller budget) is fixed—taking the smallest selected budget as binding—before any learned method is compared. The grid is coarse, so realized anchor success can fall outside the target band (0.05–0.75 at the dynamic binding budgets at calibration time, 0.25–0.62 across formal static cells; the full grid, per-suite selections, and realized rates are tabulated in the appendix). In the static experiments, calibration and evaluation share worlds (anchor-only, never touching learned outcomes; disclosed), so we describe those runs as calibrated benchmark evaluations. The dynamic 50-map cohort and the 144-map restricted-observation cohort are *confirmation* cohorts: their budgets were frozen before the cohorts were generated. Success means reaching the goal within the budget; the appendix reports exact success-versus-budget curves up to $4\times$ binding, removing dependence on any single threshold. Search effort is the *matched-solved expansion ratio*: the median over jointly solved maps of the per-map ratio of learned to anchor expansions. Because it conditions on joint success, it can favor a method that solves only easier maps; success is therefore always reported first, and empirical path-cost ratios are reported separately for the principal additive comparisons (dynamic per-suite means 1.008–1.164; appendix).

**Statistics.** Success comparisons use paired McNemar tests with Benjamini–Hochberg correction within each experiment's family of suite-level comparisons (adjusted values reported as $q$). Interval estimates are percentile bootstraps with 10,000 resamples; the resampling unit is the evaluation map, and repeated adaptation or model seeds are averaged within maps before resampling; quoted intervals reflect map uncertainty conditional on the evaluated seeds. Per-suite bootstrap intervals (as in Figure 1) are marginal; the corrected McNemar tests are primary (full policy, including declared multiplicity families, in the appendix). Method-versus-method claims use direct map-paired contrasts. Unless noted, continuous experiments use one primary training seed; the dynamic result adds two retrained pipelines; mission and refiner experiments use three-seed grids.

## Dynamic Zero-Shot Transfer

**Setting and fixed-model protocol.** The dynamic experiment trains on the dynamic maze, rooms, and spiral families and evaluates on all six dynamic suites—three sharing a training generator (disjoint maps) and three held out as above (dense maze, crossing, large rooms). The training run keeps 53 usable worlds of 72 candidates, 24 per family (usable: builds, roadmap connects, space–time solvable at $t$=0); their space–time tensors hold 1,129,536 node–time slots, of which 821,850 (72.8%) carry finite backward space–time values and are supervised (mean 15,507 per world; appendix). The original 20-map-per-suite cohort served as *development* data: based on it, we selected the field U-Net *blind* twin—the model that receives no future-motion input—and froze its checkpoint, additive integration, and per-suite budgets; the 50-map-per-suite *confirmation* cohort was then generated from a pre-specified seed range, map-disjoint by fingerprint check, with no further choices. Selection still happened, on development data; the protocol protects only the confirmation cohort. The blind heuristic receives only current-time occupancy channels; the space–time planner and collision checker always use the true deterministic trajectories when validating $(v, t)$ transitions, so the aware/blind ablation changes only the heuristic's input.

**Confirmation result.** On the confirmation cohort, the fixed blind U-Net raises success on all six suites: deltas $+0.18$ to $+0.84$, with every paired map-level 95% CI excluding zero (marginal intervals; exact McNemar tests give BH-adjusted $q \leq 0.0039$ on all six suites, and every discordant map favors the learned heuristic), and matched expansion-ratio medians of 0.047–0.273—a 73–95% reduction in search effort on jointly solved maps (Figure 1; jointly solved counts 3–41 per suite). Empirical path-cost ratios on solved maps stay within per-suite means of 1.008–1.164, and the pattern reproduces the disjoint development cohort's (appendix). Gains on the three held-out suites (dense maze $+0.64$, crossing $+0.80$, large rooms $+0.18$) establish transfer to a distinct topology and to parameter/scale shifts of trained generators; the other three establish map-level generalization within trained families.

**Independent retraining robustness.** Two additional full training pipelines with independent seeds—run, we disclose, at a larger world-collection budget than the canonical pipeline (139 and 149 usable training worlds versus 53; appendix), so they vary seed and data scale together—evaluated on the same held-out cohort, reproduce the success pattern: across the three pipelines, 17 of 18 pipeline–suite paired CIs exclude zero (deltas $+0.10$ to $+0.88$; the one interval that includes zero is seed 2001's near-ceiling large-rooms cell, where the anchor already solves 0.82). Effort magnitudes are stable across pipelines on three suites (medians 0.047–0.135) and pipeline-variable on the other three (0.216–0.625); we therefore quote effort as across-pipeline ranges on those suites. We distinguish this retraining robustness from the evaluation-cohort replication above: the retrained models are evaluated on the same 50 maps.

**Tuned weighted A* control.** Because the anchor is deliberately simple, we also compare against weighted A* with the anchor inflated to $w_h h_0$, tuned per suite over $w_h \in \{1.1, 1.2, 1.5, 2, 3, 5\}$ on the development cohort (rule frozen in advance: highest success, ties smaller) and evaluated once on the confirmation cohort; the control was specified after the learned confirmation results were known, with tuning touching only development data (appendix). The comparison yields a three-way tradeoff rather than dominance (Table 1). *Success:* no significant difference is detected on five suites—differences of at most one map on four, and the four-map weighted-A* advantage on the open crossing arena is not significant under the paper's exact McNemar policy ($q$=0.375)—while the learned heuristic holds the one BH-significant advantage, on spiral ($+0.30$ $[+0.18, +0.42]$; 15/0 discordant maps, $q$=$3.7 \times 10^{-4}$). *Effort:* learned-to-weighted expansion medians are below parity everywhere, with the marginal 95% interval excluding parity on four of six suites (0.217–0.728); the crossing and dense-maze intervals include parity, and these intervals are descriptive—the corrected success tests carry the paper's inferential policy. *Path quality:* on jointly solved maps, the additive learned heuristic pays a significant premium over tuned inflation on crossing (paired per-map difference $+0.156$ $[+0.116, +0.200]$) and large rooms ($+0.080$ $[+0.058, +0.105]$); the other four suites show no significant difference (within $\pm 0.005$). Full per-suite intervals are in the appendix.

**Safe-interval planning solves this substrate outright.** We also implemented discrete-time SIPP (Phillips and Likhachev 2011) on the identical substrate with the same collision predicates and the anchor heuristic, gated by a hard correctness check: on all 300 confirmation instances, SIPP's earliest arrival exactly equals the space–time optimum. SIPP solves 0.98–1.00 of every suite at optimal path quality, its interval-state expansions (a different unit from $(v, t)$ expansions, never merged) stay below every binding threshold (medians 76–329), and its wall time is 1.0–1.7 s per map—comparable to the learned pipeline, slower than tuned weighted A*. The learned contribution is therefore located precisely: it makes budgeted time-expanded A* effective and more expansion-efficient than tuned inflation, but it does not outperform a dedicated dynamic planner on this deterministic substrate (table in the appendix).

**External-geometry boundary.** To test the claim beyond our own generators, six MovingAI maps (Sturtevant 2012) (three street, three dungeon) were converted to the canonical substrate (64×64 occupancy, rectangle-decomposed obstacles), given the trained maze-family patroller dynamics, and evaluated with the same frozen heuristic under the frozen calibration/tuning protocol (10 development + 25 evaluation instances per group; appendix). The success advantage does not transfer: the learned heuristic never significantly beats the anchor (deltas $-0.08$ on both groups) and loses to tuned weighted A* on dungeon maps ($-0.28$, 0/7 discordant, exact $p$=0.016) while expanding more nodes there (matched ratio 2.20 $[1.25, 6.36]$) and paying significant path-quality premiums on both groups; only the street-map matched-effort advantage over the anchor survives (0.48 $[0.28, 0.64]$). The dynamic zero-shot result is therefore bounded to the procedural families and their tested shifts.

**Future-window inputs provide no consistent benefit.** The primary model is already the blind twin; the twin study quantifies the future-window contrast directly. Aware and blind twins differ in exactly one input: eight channels of ground-truth future occupancy. On the held-out cohort the contrast is suite-dependent in both directions and unstable across pipelines: of 18 contrasts, two significantly favor the aware twin, two the blind twin, fourteen neither, and the significant cells fall in different suites for different pipelines—large rooms flips from significantly blind-

| Suite | $w_h$ | Success learned | Success WA* | Expansions learned/WA* [95% CI] | Cost learned | Cost WA* |
|---|---|---|---|---|---|---|
| Crossing | 1.5 | 0.92 | 1.00 | 0.825 [0.531, 1.167] | 1.164 | 1.008 |
| Maze | 5 | 0.96 | 0.96 | 0.601 [0.440, 0.835] | 1.013 | 1.017 |
| Dense maze | 5 | 0.70 | 0.70 | 0.982 [0.823, 1.089] | 1.008 | 1.006 |
| Rooms | 3 | 1.00 | 0.98 | 0.554 [0.439, 0.798] | 1.019 | 1.015 |
| Large rooms | 1.5 | 1.00 | 0.98 | 0.728 [0.536, 0.966] | 1.083 | 1.003 |
| Spiral | 5 | **1.00** | 0.70 | 0.217 [0.183, 0.312] | 1.014 | 1.013 |

Table 1: Fixed learned heuristic versus per-suite tuned weighted A*, confirmation cohort at binding budgets. Expansions: median learned-to-weighted ratio on jointly solved maps [map-bootstrap 95% CI]. Cost: mean arrival over optimal arrival on the same jointly solved maps; the paired difference is significant on crossing ($+0.156$ [$+0.116, +0.200$]) and large rooms ($+0.080$ [$+0.058, +0.105$]), within $\pm 0.005$ elsewhere. Bold: the only BH-significant success difference (exact McNemar $q$=$3.7 \times 10^{-4}$; crossing's four-map gap $q$=$0.375$).

better ($-0.200$ [$-0.320, -0.100$]) to significantly aware-better ($+0.080$ [$+0.020, +0.160$]); the canonical pipeline also trained on fewer worlds, but the two collection-matched retrains still disagree (Figure 1(c); tables in the appendix). The original 24 suite-by-backbone cells point the same way, and after full fine-tuning at $K$=16, eight of nine success differences are non-positive (the ninth includes zero). We report a failure to observe a systematic future-window benefit under deterministic dynamics—not equivalence, and not harm; stochastic motion is untested.

## Effects of Supervision Target and Planner Integration

**Static family transfer.** On the six calibrated static suites, the transferred field HRM solves 0.75–1.00 of test maps against anchor success of 0.25–0.58, with BH-adjusted $q$ values ranging from below 0.001 to 0.025 on four suites and matched expansion ratios of 0.521–0.850; three of the six suites (dense maze, bugtrap, large rooms) are held out from training as above. Pooled models trained on the three source families also improve success on every held-out family before any target labels are used.

**Supervised target.** On the calibrated hard static maps, a pooled ON-LSTM scalar residual model raises hard-maze success from 0.525 to 1.000 over the anchor ($q$ = $4.3 \times 10^{-11}$), while the parameter-matched scalar HRM collapses to a constant prediction at the residual cap and merely matches the baseline (lower learning rates and a soft cap reproduce the collapse). Retaining the HRM while changing input representation, supervised target, and output head to the raster residual field raises hard-maze success from 0.625 to 0.975 ($q = 3.4 \times 10^{-4}$); a later experiment shows a well-trained scalar HRM is also viable (ratios 0.427–0.822). This localizes the failure to the scalar training formulation, though the field intervention changes several coupled components.

**Planner integration.** The preferred integration differs between the two tested substrates. On continuous roadmaps, additive residuals exploit the gap the Euclidean anchor leaves: field-HRM ratios are 0.521–0.850 at empirical path-cost ratios of roughly 1.02–1.14, while rank-only variants stay closer to plain anchor search. On discrete grids the learned signal is demonstrably informative—the pooled residual correlates 0.987–0.994 with the true residual across map sizes—but its magnitude is scale-flat (predicted-to-true mean falls from 0.73 to 0.19 as maps grow), and every additive configuration in the controlled 22-suite run is at or below its matched baseline. The key control reuses the *same* trained predictions only to rank expansions inside a $w$=1.0 focal band: expansions fall 6–15% with no observed success regression (Chrestien et al. 2023)—changing only the integration recovered the ordering signal. We interpret the cross-substrate pattern—additive helping against a looser anchor, ranking against a tighter one—as a hypothesis supported by these results; anchor tightness itself is not directly measured.

**Oracle integration control.** An oracle control removes model-estimation error: the same exact value ranker loses all six development-map comparisons when paired with a fresh certifier search that discards the incumbent's work (141.2 versus 129.7 mean expansions) and wins all six when sharing one search queue and $g$-state (122.8)—identical values, only search-state reuse changed; a development mechanism test, not a general interface claim.

## Few-Shot Adaptation

**Direct contrasts: LoRA protects at $K$=1; full fine-tuning overtakes on effort by $K$=16.** At $K$=1 the map-paired full-FT−LoRA success difference is negative in all six target/backbone cells and significant in four (worst $-0.600$ [$-0.773, -0.407$] on large rooms/HRM); LoRA retains the zero-shot gain (dense maze ratio 0.686 [$0.590, 0.774$], success $+0.467$) while full fine-tuning reaches ratio 1.293 on large rooms and scratch is significantly harmful on dense maze ($-0.167$). At $K$=16 the contrast reverses on effort: the paired expansion-ratio difference favors full fine-tuning in all six cells, significantly in five (up to $-0.366$ [$-0.489, -0.235$]), at near-parity success (one cell significantly favors full fine-tuning; marginal intervals agree, e.g. dense maze 0.610 versus 0.786, disjoint). A 27-cell matched-compute ablation finds bounded and unbounded LoRA nearly identical (median map-level ratio delta 0.0000, maximum 0.037), so the output clamp is not the primary cause of the LoRA plateau; LoRA is not universally protective either (bugtrap ratio 1.153 [$1.012, 1.523$] by $K$=32). Full per-cell contrasts and illustrative $K$-indexed curves are in the appendix.

**Dynamic maps supply many labels per map.** The dy-

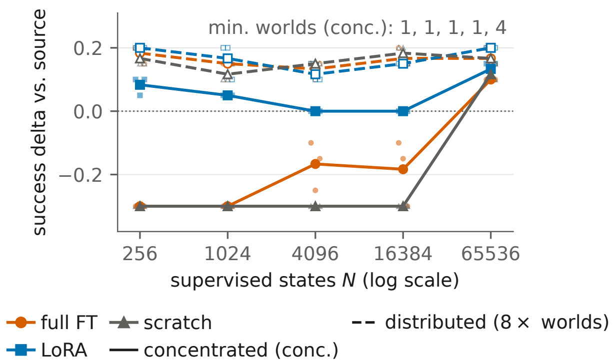


Figure 2: The matched-label factorial, dynamic substrate (static panel and per-cell tables in the appendix): success delta versus the frozen zero-shot source at fixed supervised-state count $N$, frozen test cohort, binding budgets. Solid: concentrated supervision (fewest worlds supplying $N$; minimum counts annotated); dashed: the same $N$ over $8\times$ as many worlds; small markers are the three adaptation seeds (each cell uses one sampled world set). Concentrated supervision causes a held-out success drop for full fine-tuning and scratch; distributing the same label count prevents it; recovery at $N$=65,536 coincides with the minimum world count rising to 4. Independent world-draw replicates reproduce the coverage contrasts; one finds a significantly degraded LoRA cell (appendix).

namic training run supervises 821,850 reachable $(v, t)$ states from 53 worlds—15,507 labeled states per world on average (collection arithmetic in the appendix). At dynamic $K$=1, full fine-tuning is not systematically catastrophic and LoRA continues to improve rather than merely preserve the base (appendix). Scratch models can post attractive solved-only ratios while solving fewer maps, so success leads those comparisons.

**A matched-label factorial separates label count from distinct-world coverage.** The $K$-indexed protocol confounds two quantities: adding maps adds labels and worlds together. A preregistered factorial therefore holds the number of supervised states $N$ fixed ($N \in \{256, 1024, 4096, 16384, 65536\}$) and varies only whether those states are drawn from the fewest worlds that can supply them (*concentrated*) or from eight times as many (*distributed*), in both domains, under matched optimizer steps (Figure 2; protocol, tables, and regression in the appendix). Each cell adapts one deterministically sampled world set—the three seeds vary initialization and batch order only—so replication over independent world draws is reported separately in the appendix. Under matched optimizer steps and within the tested range, state count did not explain the observed success degradation. Concentrated supervision causes a held-out success drop for full fine-tuning and scratch in both domains—−0.300 in every seed at dynamic concentrated $N \leq 1{,}024$, still negative in every seed at $N$=16,384 from a single world (significantly in one of three)—while distributing the *same* states across more worlds prevents the drop (+0.15 to +0.20; Figure 2). No statistically significant low-rank-adaptation success degradation appears in any of the original 60 cells (worst $-0.067$ $[-0.167, 0.000]$); absent a noninferiority margin this is a failure to observe degradation, not an equivalence claim, and independent world-draw replicates make the caveat concrete: the coverage contrasts reproduce (distributed−concentrated +0.13 to +0.50 for full fine-tuning and scratch, all CIs excluding zero), but one replicate LoRA cell does degrade significantly (static concentrated $N$=256: $-0.133$ $[-0.267, -0.033]$, still the least-degraded method in that cell; appendix). On the static matched expansion ratios (the domain with adequate jointly solved counts), full fine-tuning is at or below LoRA at every tested $N$ and the full-fine-tuning$\times \log N$ interaction is null ($-0.0025$ $[-0.0087, +0.0028]$); dynamic effort cells contain one jointly solved map and support no inference, and because ratios condition on solved maps, degraded-success cells can look spuriously efficient—so success is the factorial's primary endpoint. The protection low-rank adaptation provides at low supervision appears in *success* and tracks distinct-world coverage rather than label count in this factorial.

## Scope and Boundary Results

**Discrete transfer.** The pooled HRM base improves mean grid success 0.591→0.612 with lower mean expansions (descriptive); adapters do not improve broadly; a 3–8-seed rank-only pilot on selected maps finds 6–15% expansion reductions at $w$=1.0 with equal or higher observed success (no full-suite confirmation).

**Composition, architecture, and hierarchical depth.** Descriptor-weighted composition of 16 single-family adapters routes to the correct family (own-axis mass ≥0.986) yet never significantly beats the pooled zero-shot model (appendix) (Wortsman et al. 2022; Ilharco et al. 2023). Architecture rankings differ across tasks and input representations (the U-Net is strongest in the dynamic field study; the HRM holds the lowest single-suite dynamic ratio; the ON-LSTM leads the early calibrated scalar stage). In compositional multi-leg missions with verified oracle headroom (three-seed grids; appendix), no structured model beats the MLP at two depths with a non-decreasing gap, and learned halting moves *opposite* to the depth hypothesis (map-level $\rho = -0.578$, stratified permutation $p = 5 \times 10^{-5}$); five of 33 recurrent cells suffer constant-prediction optimization collapse and are excluded from capacity conclusions. Follow-ups on persistent memory, iterative refinement (Tamar et al. 2016; Lee et al. 2018), hierarchical substitution, and persistent planning-time search state are also negative (appendix). These are formulation-specific negatives, not claims that recurrence, memory, or hierarchy are universally ineffective (Jolicoeur-Martineau 2025; Czechowski et al. 2021).

**Restricted-observation transfer.** A final static study restricts the learned model to a radius-0.20 local subgraph, trains it from behavior-derived returns rather than shortest-path labels, and integrates it through one inference-time Bellman backup over that subgraph. Under a frozen comparison on a disjoint 144-map cohort—whose path-quality gate is the comparator-relative criterion adopted after the original absolute ceiling failed in development (chronology in the appendix)—this local MLP reduces pooled expansions against the complete-map field HRM by 12.96 (95% CI $[-16.30, -9.74]$; 109/3/32 wins/ties/losses) at better empirical path quality (cost ratios 1.024/1.162 versus 1.031/1.335), and a separate $w$=1.10 bounded control passes all 144 suboptimality certificates. Sequential ablations locate the sign change at the backup (static insertion loses by $+16$ expansions; one backup flips the point estimate to $-1.21$ with an interval crossing zero; the frozen scale calibration then yields the confirmed reduction). The study is static, uses one model seed and cohort, carries no formal bound on the direct arm, and is slower in wall time (tables in the appendix).

## Related Work

**Learned guidance for search.** Learned heuristics have been trained by bootstrapping (Arfaee, Zilles, and Holte 2011), oracle imitation (Bhardwaj, Choudhury, and Scherer 2017), differentiable search (Yonetani et al. 2021), transformer grid guidance (Kirilenko et al. 2023), classical-planning objectives (Ferber, Helmert, and Hoffmann 2020), and ranking objectives (Chrestien et al. 2023). We retain classical graph search throughout, evaluate cross-family and dynamic transfer of identical trained models, and compare additive against rank-only integration of the *same* predictions (Pohl 1970; Pearl and Kim 1982; Aine et al. 2014); to our knowledge, prior work has not evaluated these factors jointly.

**Dynamic, real-time, and motion planning.** The dynamic substrate uses time-expanded space–time search in the spirit of cooperative A* (Silver 2005); safe-interval planning (Phillips and Likhachev 2011) is a related, typically faster alternative that we do not evaluate. The restricted-observation study is related to real-time heuristic search (Korf 1990) and Real-Time Adaptive A* (Koenig and Likhachev 2006); its one-step backup resembles a single value-iteration-network update (Tamar et al. 2016; Lee et al. 2018) and neural algorithm execution (Veličković and Blundell 2021). The backup here runs once, at inference, over a radius-bounded subgraph, purely to integrate a transferred prediction. Learned samplers and end-to-end motion planners (Ichter, Harrison, and Pavone 2018; Qureshi et al. 2021) replace the planner itself; we keep a fixed classical planner and learn only its guidance, making heuristic transfer measurable on a PRM (Kavraki et al. 1996).

**Adaptation and recursive architectures.** LoRA (Hu et al. 2022), model soups (Wortsman et al. 2022), and task arithmetic (Ilharco et al. 2023) motivate the low-rank and composition studies; the contribution is not the use of LoRA but the comparison of low-rank against full-rank updates as a function of label availability and, through the factorial, of how supervision is distributed across worlds. HRM (Wang et al. 2025), adaptive computation (Graves 2016; Banino, Balaguer, and Blundell 2021), and simplified recursive models (Jolicoeur-Martineau 2025) motivate the architecture experiments; our results concern heuristic regression and integration, not their benchmark claims.

## Limitations

Most continuous primary results are single-GPU validations with one primary training seed, and several experiments reuse trained bases or test maps rather than independent replications. Only the 50-map cohort is selection-free; static calibration shares worlds with evaluation (anchor-only; budget curves mitigate); baselines are the anchor, tuned weighted A*, and safe-interval planning—external *learned* planners are not compared. The smallest jointly solved cell is three maps; matched-effort magnitudes vary by pipeline on three suites; a best-configuration view in the appendix is post-hoc. The retrained pipelines used larger collections (139/149 versus 53 worlds; comparisons vary seed and scale together); the expansion advantage over weighted A* excludes parity on four of six suites and costs path quality on two. Quoted intervals underrepresent training-run uncertainty (addressed only dynamically, via across-pipeline ranges); the discrete result covers selected cells; map-clustered reanalysis supports the quoted conclusions, though not every exploratory table was regenerated and no equivalence tests were run. The factorial covers one target family per domain, three adaptation seeds, and one sampled world set per cell (independent-draw replicates in the appendix); coverage counts worlds rather than measuring geometric diversity; matching covers labels and steps, not world-generation cost; per-cell readouts carry no multiplicity adjustment or noninferiority test. The primary dynamic experiments report expansions rather than end-to-end latency (timing decompositions in the appendix).

The restricted-observation confirmation is static, one seed and cohort, and tests input restriction only; the planner still knows the roadmap; the direct arm has no formal bound; both operating points are slower in wall time despite fewer expansions. Dynamic obstacles are deterministic; the work does not cover stochastic prediction, kinodynamic constraints, multi-agent coupling, physical robots, or environments beyond 2-D single-agent navigation. The MovingAI evaluation bounds external validity: the zero-shot claims are specific to the procedural substrate and its tested shifts. Negative results are failures to observe improvement under pre-specified protocols, not universal rejections; safety-critical deployment would additionally require uncertainty-aware motion models, collision guarantees, and end-to-end validation.

## Conclusion

Across the roadmap suites and a discrete rank-only pilot, learned heuristics generalized beyond their training maps: the frozen blind U-Net improved success on all six dynamic suites while cutting matched effort 73–95% against the anchor and staying more expansion-efficient than tuned weighted A* on four of six suites. Safe-interval planning solves the substrate outright and transfer does not reach externally authored geometry, bounding the contribution to search-effort structure within budgeted search. Target and interface can each reverse a conclusion; low-rank adaptation protects under scarce supervision, full fine-tuning overtakes as supervision spans more worlds, and the factorial ties failure to too few distinct worlds; evaluate learned heuristics with search interface, path quality, and supervision distribution.

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